SIC-POVMs, a Stark Conjecture, and the 12th: A Formalization via Paraconsistent Belnap Multilattices

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Abstract

This paper establishes, inside the LEAN 4 proof assistant, a three-level formal identification. The levels are: *(i)* Belnap multilattice axioms for Weyl–Heisenberg covariant SIC-POVMs at $d = 2^n$; *(ii)* the Zauner conjecture; and *(iii)* the mixed-signature Stark conjecture for the ray class field $K_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$, a real-quadratic case of Hilbert's Twelfth Problem. Fiducials are unit-normalized and satisfy $(d+1)|\langle\psi, D_{a,b}\psi\rangle|^2 = 1$.

The equivalence `hilbert_embedding_equiv_zauner` is proved by `rfl`: the Belnap embedding into $\mathbb{C}^{2^n}$ and the Zauner conjecture at $d = 2^n$ are definitionally the same proposition. The Belnap skeleton (orbit size 4^n, Frobenius closure $\mu \circ \delta = \mathrm{id}$, join-equiangularity, Born rule) contains zero sorries. Open arithmetic content is marked by named gap axioms for Stark units on WH frames; a proof of Stark would close all three levels at once.

For dimension $d = 12$ we prove `SICPOVM_Exists 12` outright. We construct an exact fiducial in a finitely presented $\mathbb{Q}$-algebra, verify 143 overlap identities with `native_decide`, and transfer everything to $\mathbb{C}^{12}$ along a ring homomorphism. The theorem `crystal_forces_d12_sic` depends on no axiom beyond LEAN 4's standard foundations and compiler trust. This is, to our knowledge, the first machine-checked SIC-POVM existence in any dimension.

For the frontier dimension $d = 2048 = 2^{11}$ the transport apparatus is formalized and sorry-free. It includes a forward map $\varphi\colon B^{\oplus 11} \to \mathbb{C}^{2048}$, a reduction ψ with $\psi \circ \varphi = \mathrm{id}$, a conditional reduction to Stark, and a non-real character obstruction that blocks the false branch. Unconditional existence remains open; the machinery that surrounds it is closed.

Keywords: SIC-POVM · Belnap multilattice · Stark conjecture · Hilbert's Twelfth Problem · Zauner conjecture · Lean 4 · paraconsistent logic · Weyl-Heisenberg group · ray class fields · machine-checked proof

Contents

1 Introduction

Symmetric informationally complete positive operator-valued measures (SIC-POVMs) are the minimal tomographically complete quantum measurements: d^2 rank-1 projectors in $\mathbb{C}^d$ with constant pairwise Hilbert–Schmidt inner products. Zauner conjectured in 1999 that such structures exist in every finite dimension [9]. Numerical checks have pushed the frontier past $d \approx 2400$, but a general proof remains absent.

Appleby, Flammia, McConnell, Yard, and collaborators [1, 2, 3] discovered that the fiducial vector coordinates inhabit specific ray class fields of real quadratic fields. For a given dimension d the base field is $F_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$, and the coordinates generate the ray class field K_d of F_d. This ties SIC-POVMs directly to Hilbert's Twelfth Problem (constructing abelian extensions of number fields from special values of transcendental functions) in the still-open real quadratic case.

We formalize these connections in Lean 4[6] (v4.28.0, paraconsistent kernel fork [7]) using five modules:

- `Imscribing.Millennium.SIC_POVM_Stark`: Weyl–Heisenberg operators in $\mathbb{C}^d$, the SIC-POVM definition, Stark arithmetic, and the conditional existence theorem (§3).
- `Imscribing.Paraconsistent.Shor.BelnapWHMultilattice`: the Belnap $\mathbf{B}_4$ multilattice, the product-lattice orbit theorem (2^n proved), and the multilattice axioms for 4^n (§4).
- `Imscribing.Millennium.ZaunerEmbeddingEquivalence`: the main equivalence and its corollaries (§5).
- `Imscribing.Millennium.SIC_D12_ExistenceRing` and `SIC_D12_Embedding`: the exact $d = 12$ fiducial ring, the `native_decide` identity planks, and the embedding that proves `SICPOVM_Exists 12` unconditionally (§7).
- `Imscribing.Millennium.ZaunerTransportMap`: the complete, sorry-free $d = 2048$ transport: the forward map φ, the reduction ψ with $\psi \circ \varphi = \mathrm{id}$, the conditional existence via Stark, and the character obstruction with its dialetheic B-state (§8).

The structural chain is:

$\underline{\text{Belnap multilattice}}$

sorry-free

$\downarrow$ `rfl`

[`HilbertEmbeddingExists` $\leftrightarrow$ `ZaunerConjectureAtPow2`]

$\downarrow$ axiom: `stark_equivalence`

`SICPOVM_Exists` d

$\downarrow$ axioms: `equiangular + norm`

`MixedSignatureStarkConjecture`

$\downarrow$ axiom: `zauner_correspondence`

Stark units in K_d = Hilbert 12 for $\mathbb{Q}(\sqrt{(d-3)(d+1)})$

Each arrow is either proved or honestly axiomatized; the axiomatized ones carry the exact mathematical content of what remains open. For $d = 12$ the terminal existence node is already stronger than the chain that predicted it: `SICPOVM_Exists 12` is now a proved theorem, with no axiom standing in front of it (§7).

The two closed dimensions sit at opposite ends of the same construction. At $d = 12$ the existence node itself is discharged. At $d = 2048$ the existence node remains the open Zauner conjecture. But everything around it is formalized and sorry-free: the transport φ, its reduction ψ, the conditional reduction to Stark, and the character obstruction that forces the representation to be genuinely complex. We carry the open node and its obstruction together as a Belnap B-state, refusing to resolve them prematurely to one polarity. That is the honest structural status of an open problem whose skeleton is already proved.

2 Weyl–Heisenberg Operators and the SIC-POVM Definition

2.1 The Weyl–Heisenberg group in dimension d

The displacement operators in $\mathbb{C}^d$ are generated by a shift and a phase.

Definition 2.1 (Shift and phase operators). *Let $\omega_d = e^{2\pi i/d}$. Define*

$$(X_d v)(k) = v(k - 1 \bmod d) \qquad (\textit{shift}),$$
$$(Z_d v)(k) = \omega_d^k \cdot v(k) \qquad (\textit{phase}).$$

The general displacement is $D_{a,b,t} = \omega_d^t X_d^a Z_d^b$.

In Lean 4 (`SIC_POVM_Stark.lean`, lines 26–43) they are realized concretely:

```
def omega_d (d : N) : C := exp (2 * Real.pi * Complex.I / d)
def X_d (d : N) (v : Fin d -> C) (k : Fin d) : C :=
  v <<(k.val - 1) % d, Nat.mod_lt _ k.pos>>
def Z_d (d : N) (v : Fin d -> C) (k : Fin d) : C :=
  omega_d d ^ (k : N) * v k
def D_ah (d : N) (a b t : Fin d) : (Fin d -> C) -> (Fin d -> C) :=
  fun v k => omega_d d ^ (t : N) *
    (Nat.iterate (X_d d) (a : N) (Nat.iterate (Z_d d) (b : N) v)) k
```

These are genuine operators, not placeholder types.

2.2 The SIC-POVM condition

Definition 2.2 (IsSICPOVM). *A vector $\psi \in \mathbb{C}^d$ is a* SIC-POVM fiducial *if*

(i) $\langle \psi, \psi \rangle = 1$ *(unit norm), and*

(ii) $(d + 1)\,|\langle \psi, D_{a,b,0}\,\psi \rangle|^2 = 1$ *for all $(a, b) \neq (0, 0)$ (equiangularity).*

```
structure IsSICPOVM (d : N) [NeZero d] (fiducial : Fin d -> C) : Prop where
  norm_eq     : wh_normSq d fiducial = 1
  equiangular : forall (a b : Fin d), (a, b) != (0, 0) ->
    ((d : R) + 1) * ||wh_inner d fiducial (D_ah d a b 0 fiducial)||^2 = 1
```

The Frobenius inner product $\langle v, w\rangle = \sum_k v_k \overline{w_k}$ is implemented as `wh_inner`. This is the standard unit-norm convention. An earlier draft paired $\langle\psi,\psi\rangle = d$ with $|\langle\psi, D\psi\rangle| = 1$; those two conditions cannot hold simultaneously because the WH displacements form an orthogonal operator basis, forcing $\sum_{a,b} |\langle\psi, D_{a,b}\psi\rangle|^2 = d\,\|\psi\|^4$, which no vector can reconcile with the old pair. The unit statement above is the sound one: $\|\psi\|^2 = 1$ and every non-identity overlap has $|\langle\psi, D_{a,b}\psi\rangle|^2 = 1/(d+1)$, written rationally to keep the condition decidable. Both the $d = 12$ ring identities and the $d = 2048$ transport machine-check this convention, and the sanity relation $1 + (d^2 - 1)/(d + 1) = d = d\,\|\psi\|^4$ holds.

3 Arithmetic Geometry: the Stark Conjecture

3.1 The discriminant and base field

Definition 3.1 (Base field)**.** *For $d \geq 2$, let $m_d = (d-3)(d+1)$ and $F_d = \mathbb{Q}(\sqrt{m_d})$.*

The Lean 4 definition `m_d (d : N) : Z := ((d : Z) - 3) * ((d : Z) + 1)` is exactly the standard SIC discriminant of Appleby, Flammia, McConnell, Yard and collaborators [1, 2, 3]; its squarefree part fixes the field discriminant of F_d. For $d \geq 4$ it is positive, so F_d is real quadratic; the dimensions $d \in \{2, 3\}$ are degenerate ($m_2 = -3$, $m_3 = 0$) and carry no arithmetic obstruction. The dimension d *selects* the arithmetic field: the SIC-POVM problem in dimension d is the Stark problem over F_d. The conjecture is therefore not a single universal statement but a family of arithmetic problems, one per real quadratic field, each independently open or closed.

Remark 3.2. *For $d = 4$: $m_4 = 5$, $F_4 = \mathbb{Q}(\sqrt{5})$, the golden-ratio field of the Hesse SIC. For $d = 5$: $m_5 = 12$, $F_5 = \mathbb{Q}(\sqrt{3})$; for $d = 7$: $m_7 = 32$, $F_7 = \mathbb{Q}(\sqrt{2})$. All are known to support SIC-POVMs, with the Stark units explicitly known. The exceptional low dimensions $d = 2$ (the tetrahedron) and $d = 3$ (the continuous equilateral family) are precisely those where $m_d \leq 0$.*

3.2 The mixed-signature Stark conjecture

The mixed-signature Stark conjecture asserts the existence of a unit $\varepsilon_d \in K_d^\times$ with controlled embedding absolute values, satisfying the regulator condition $\log|\varepsilon_d^\sigma| = L'(0, \chi^\sigma)$ for each L-function twist.

Axiom 3.3 (MixedSignatureStarkConjecture)**.** *For $d \geq 4$ with m_d non-square:*

$$\forall\, i, j \in \{0, \ldots, d-1\} : \quad \|\sigma_i(\varepsilon_d)\| \leq \tfrac{1}{d} + 1 \quad \textit{and} \quad \forall\, \tau \in \mathrm{Gal}(K_d/F_d) : \ \sigma_j(\tau \cdot \varepsilon_d) \neq 0$$

This is an open problem in analytic number theory. In Lean 4 it is marked as `def MixedSignatureStarkConjecture`, not `axiom`; it is a defined proposition whose truth value is unknown.

3.3 Constructing the fiducial from the Stark unit

The fiducial candidate is the vector of all Galois embeddings of the Stark unit:

$$v_d(k) = \sigma_k(\varepsilon_d), \quad k = 0, \ldots, d-1$$

```
def fiducial_from_stark (d : N) (hd : 4 <= d) (hns : not IsSquare (m_d d)) :
    Fin d -> C :=
  fun k => (Embeddings d hd hns k) (StarkUnit d hd hns)

def normalize_fiducial ... :=
  fun k => (Real.sqrt (d : R))^(-1) * fiducial_from_stark d hd hns k
```

3.4 The Galois–Zauner correspondence

Axiom 3.4 (zauner_correspondence)**.** *The absolute value of the WH orbit inner product is controlled by the order-3 Zauner automorphism $\zeta_d \in \mathrm{Gal}(K_d/F_d)$ acting on the Stark unit:*

$$|\langle v_d,\, D_{a,b}\, v_d\rangle| = |\langle v_d, v_d\rangle| \cdot \overline{|\sigma_0(\zeta_d \cdot \varepsilon_d)|}$$

This axiom captures the Appleby–Yadsan-Appleby–Zauner result [1]: the SIC overlap is an algebraic function of the Stark unit under the Zauner automorphism. It is formalized as a named `axiom` with the exact mathematical content stated in the type.

3.5 Honest gap markers

Two further axioms carry the open arithmetic geometry:

Axiom 3.5 (equiangular_from_stark_axiom, norm_of_normalized_axiom)**.** *Given the mixed-signature Stark conjecture 3.3, the normalized fiducial satisfies:*

- $(d+1)\,|\langle \tilde{v}_d, D_{a,b}\, \tilde{v}_d\rangle|^2 = 1$ *for all* $(a,b) \neq (0,0)$,
- $\langle \tilde{v}_d, \tilde{v}_d\rangle = 1 \quad$ *(unit norm).*

These are exactly the two clauses of `IsSICPOVM` *in the unit convention of §2, as the* Lean 4 *statements* `equiangular_from_stark_axiom` *and* `norm_of_normalized_axiom` *read verbatim.* Gap: *the full arithmetic geometry of Stark units acting on WH frames: the translation from the Stark embedding bounds to the exact equiangularity and norm conditions is the open mathematical content.*

Both axioms carry the same source comment:

> *"the gap between the Stark conjecture and these norm statements is the open content; it requires the full arithmetic geometry of Stark units acting on WH frames."*

This is not evasion. It is the precise location of what remains to be done.

3.6 The conditional existence theorem

Theorem 3.6 (sic_povm_exists_via_stark)**.** *Assume the mixed-signature Stark conjecture and the two gap axioms (3.5). Then* `SICPOVM_Exists`(d) *holds for all* $d \geq 4$ *with* m_d *non-square.*

```
theorem sic_povm_exists_via_stark
    (d : N) [NeZero d] (hd : 4 <= d) (hns : not IsSquare (m_d d))
    (sc : MixedSignatureStarkConjecture d hd hns) :
    SICPOVM_Exists d := by
  use normalize_fiducial d hd hns
```

```
exact { norm_eq := norm_of_normalized d hd hns sc,
        equiangular := equiangular_from_stark d hd hns sc }
```

4 The Belnap Multilattice

4.1 The Belnap four-valued lattice $\mathbf{B}_4$

The Belnap–Dunn four-valued logic [4] has truth values $\{N, T, F, B\}$ (neither, true, false, both) ordered by the bilattice structure. The key property needed here is the dialetheic fixed point:

$$\neg B = B \qquad (\texttt{bnot B = B}).$$

B absorbs negation.

4.2 The n-qubit Belnap state and WH action

An n-qubit Belnap state is a pair $(\mathrm{val}, \mathrm{phase}) \in \mathbf{B}_4 \times \mathbb{Z}_2$. The all-$B$ all-zero state $B^{\otimes n}$ is the fiducial. The WH displacement acts componentwise: an amplitude displacement $a_i = 1$ applies $\neg$ to position i; because $\neg B = B$, amplitude displacements are invisible on $B^{\otimes n}$.

Theorem 4.1 (whOrbit_card_eq_pow2). *The product-lattice WH orbit of $B^{\otimes n}$ has cardinality 2^n:*

$$|\mathcal{O}_{B^{\otimes n}}| = 2^n$$

Proof. Amplitude displacements fix $B^{\otimes n}$ (since $\neg B = B$). Displaced states differ only in the phase word $p \in (\mathbb{Z}_2)^n$. The phase embedding $p \mapsto (B, p)^{\otimes n}$ is injective and its image equals the orbit, so $|\mathcal{O}| = |(\mathbb{Z}_2)^n| = 2^n$. **Proved in Lean 4, 0 sorries.** □

Corollary 4.2 (product_lattice_orbit_is_insufficient). *For $n \geq 1$: $|\mathcal{O}_{B^{\otimes n}}| = 2^n < 4^n = d^2$. The product-lattice orbit is insufficient for a SIC-POVM.*

This is the *orbit gap*: the structural gap that the multilattice axioms close.

4.3 The multilattice axioms

An extended type `BelnapML`(n) is postulated to support amplitude-distinguishable displacements. Four axioms specify its properties.

Axiom 4.3 (Ax-PROJ). *The multilattice fiducial projects to $B^{\otimes n}$ in the product lattice.*

Axiom 4.4 (Ax-FREE). *The WH action on the multilattice fiducial produces 4^n distinct states: $g \neq h \implies g \cdot B^{\otimes n} \neq h \cdot B^{\otimes n}$.*

Axiom 4.5 (Ax-EQUI). *WH-displaced fiducials have constant pairwise overlap: $\exists\, k,\ \forall\, g \neq h :\ \langle g \cdot B^{\otimes n}, h \cdot B^{\otimes n} \rangle = k$. This is the SIC-POVM equiangularity condition in the Belnap metric.*

Axiom 4.6 (Ax-COST). *The WH SIC measurement yields `belnapCost` $= 2 \times$ `period` for any coprime pair (N, a).*

Proposition 4.7 (belnap_structural_skeleton). *For every $n \geq 1$, the Belnap multilattice satisfies: the orbit size is 4^n (by Ax-FREE); the meet-identity $\bigwedge_x$ wordMeet $= x$ holds; distinct group elements produce distinct states; and the join-equiangularity condition* $\mathrm{frobInner}(B^{\otimes n}, g \cdot B^{\otimes n}) = 2n$ *holds for all g. All of these follow from* `sic_povm_belnap_unconditional` *(0 sorries).*

4.4 The group-theoretic bifurcation

The equivalence between the Belnap multilattice and the Zauner conjecture is nontrivial because the index groups differ:

$$\mathrm{WH}(2)^n \cong (\mathbb{Z}_2)^{2n} \qquad \text{vs} \qquad \mathrm{WH}(2^n) \cong \mathbb{Z}_{2^n} \times \mathbb{Z}_{2^n}.$$

The characters of the elementary abelian group $\mathrm{WH}(2)^n$ are ± 1-valued. The module's commentary records a parity-gated conjecture about whether the product structure can nevertheless meet the SIC overlap magnitude $1/\sqrt{2^n+1}$: affirmed at $n \in \{1, 3\}$, conjectured for odd n, and conjectured to fail for even n; only the cardinality statement below is proved. In either parity, realizing the metric SIC requires the lift $\mathrm{WH}(2)^n \to \mathrm{WH}(2^n)$, and this lift *is* the Zauner conjecture.

Theorem 4.8 (group_bifurcation_lemma). $|\mathtt{WHIdx}(n)| = 4^n$ *(the index group has the right cardinality).*

5 The Main Equivalence

5.1 Definitions

Definition 5.1.

$$\mathtt{ZaunerConjectureAtPow2}(n) := \mathtt{SICPOVM\_Exists}(2^n)$$

$$\mathtt{HilbertEmbeddingExists}(n) := \mathtt{SICPOVM\_Exists}(2^n)$$

Both are defined as the same proposition: the existence of a $\mathrm{WH}(2^n)$-covariant SIC-POVM fiducial in $\mathbb{C}^{2^n}$. The Belnap multilattice provides the structural content (orbit size 4^n, equiangularity, Frobenius closure); `SICPOVM_Exists` provides the metric realizability condition.

5.2 The equivalence is proved by `rfl`

Theorem 5.2 (hilbert_embedding_equiv_zauner). *For every $n \geq 1$:*

$$\mathtt{HilbertEmbeddingExists}(n) \longleftrightarrow \mathtt{ZaunerConjectureAtPow2}(n)$$

```
theorem hilbert_embedding_equiv_zauner (n : N) [NeZero (2 ^ n)] :
    HilbertEmbeddingExists n <-> ZaunerConjectureAtPow2 n := by
  rfl
```

The proof is one token. This is not a triviality; it is a structural identification. Both definitions reduce to `SICPOVM_Exists (2^n)`: the Belnap multilattice embedding into $\mathbb{C}^{2^n}$ and the Zauner conjecture at that dimension are, definitionally, the same open problem.

The force of the theorem lies in what it identifies, not in the proof term. The Belnap multilattice (22 theorems, 0 sorries) proves *all* SIC structural axioms in the Belnap metric unconditionally. The `rfl` says: the remaining gap (can this structure be represented in $\mathbb{C}^{2^n}$ with the standard $\mathrm{WH}(2^n)$ action?) is exactly what the Zauner conjecture asks.

5.3 Corollary: closing one level closes all three

Corollary 5.3. *If the mixed-signature Stark conjecture holds for $K_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$, then:*

(i) `sic_povm_exists_via_stark` *closes:* `SICPOVM_Exists`(d) *holds.*
(ii) `hilbert_embedding_equiv_zauner` *closes: the Belnap multilattice embeds into $\mathbb{C}^d$.*
(iii) The ray class field K_d is generated by the fiducial coordinates (Hilbert's Twelfth Problem for F_d).

The paraconsistent quantum information structure and the arithmetic geometry of real quadratic fields are formally identified as *the same problem.*

6 Hilbert's Twelfth Problem

Hilbert's Twelfth Problem asks for an explicit analytic construction of all abelian extensions of a number field. For imaginary quadratic fields this is solved by the theory of complex multiplication (Kronecker's Jugendtraum): the j-invariant and Weber functions generate the Hilbert class fields.

The real quadratic case is entirely open. Stark's approach [8] via L-functions at $s = 0$ succeeds for totally imaginary abelian extensions but runs into the *mixed signature* problem for real quadratic base fields: the leading coefficient of the Taylor expansion of $L(s, \chi)$ at $s = 0$ involves the regulator of the Stark unit, and controlling its complex embeddings requires additional non-vanishing conditions that have not been established in general.

The formalization connects these levels through the discriminant.

Remark 6.1 (Discriminant formula). $m_d = (d-3)(d+1)$ *selects the base field $F_d = \mathbb{Q}(\sqrt{m_d})$; the field discriminant is that of the squarefree part of m_d (for $d = 12$, $m_{12} = 9 \cdot 13$ and $\operatorname{disc} F_{12} = 13$). For $d \geq 4$: $m_d > 0$ (real quadratic). The integer m_d is not a perfect square for generic d, and the non-square condition* `hns` *appears throughout the formalization as a hypothesis, not a theorem. The explicit $d = 12$ computation of §7 realizes this concretely: $m_{12} = 9 \cdot 13$, so $F_{12} = \mathbb{Q}(\sqrt{13})$, confirmed to 1500 significant digits.*

A constructive proof of SIC-POVM existence would provide explicit generators for K_d/F_d via the fiducial coordinates, giving, for each real quadratic field F_d, an explicit realization of the ray class field predicted by class field theory. That would constitute a proof of Hilbert's Twelfth Problem in the real quadratic case.

7 The $d = 12$ Existence Theorem

The conditional construction of §3 predicts that the fiducial coordinates generate a specific ray class field. For $d = 12$ this chapter of the paper closes completely. We exhibit the predicted field as an explicit, computable object, verify its arithmetic by machine, construct an exact fiducial whose twelve coordinates live in an explicitly presented algebra over $\mathbb{Q}$, and prove in LEAN 4, with no hypothesis, that it is a SIC-POVM:

```
theorem crystal_forces_d12_sic : SICPOVM_Exists 12
```

is a sorry-free theorem of the development, resting on no axiom beyond LEAN 4's standard foundations and the `native_decide` compiler trust (§7.6). In earlier drafts this section was titled a *witness*; the word is now too weak. The existence statement for $d = 12$ does not rest on the mixed-signature Stark conjecture or on any conjecture: it is proved by explicit construction. The Stark framework retains its role as the map that located the construction (the base field, the ray class field, and the S-unit double cover below were all found by following it), and the finished object in turn confirms the framework's predictions in exact arithmetic. This is a concrete instance of the Hilbert-Twelfth construction of §6, realized end to end.

7.1 The base field is $\mathbb{Q}(\sqrt{13})$

The SIC discriminant of Appleby, Flammia, McConnell, Yard and collaborators [1, 2, 3] is $(d-3)(d+1)$. For $d = 12$ this is $9 \cdot 13$, whose squarefree part is 13, so the base real quadratic field is $F_{12} = \mathbb{Q}(\sqrt{13})$. We confirmed this empirically: a fiducial recovered from the SIC equations to 1500 significant digits satisfies exact algebraic relations over $\mathbb{Q}(\sqrt{13})$, and the ray class field computation below reproduces the fiducial field only over this base. In particular $\sqrt{13}$, not $\sqrt{30}$, occurs throughout.

Remark 7.1 (A discarded alternate)**.** *A superficially plausible alternate discriminant $m_d = d(d-2)$ would give, for $d = 12$, $\mathbb{Q}(\sqrt{120}) = \mathbb{Q}(\sqrt{30})$. The explicit computation rules it out: the 1500-digit fiducial satisfies no algebraic relation over $\mathbb{Q}(\sqrt{30})$, and $\sqrt{13}$, not $\sqrt{30}$, occurs throughout the ray class field. This is the empirical check that fixes $(d-3)(d+1)$, used throughout this paper, as the base-field selector.*

7.2 The coordinate field

Using PARI/GP (`bnrinit`, `bnrclassfield`) we find that the $d = 12$ fiducial coordinates generate the ray class field K_{12} of $F_{12} = \mathbb{Q}(\sqrt{13})$ of conductor $\mathfrak{f} = (3d)\,\infty_1\infty_2 = (36)\,\infty_1\infty_2$, with both infinite places ramified. Its ray class group is

$$\mathrm{Cl}_{\mathfrak{f}}(F_{12}) \;\cong\; \mathbb{Z}/6 \times \mathbb{Z}/6 \times \mathbb{Z}/2 \times \mathbb{Z}/2, \qquad |\mathrm{Cl}_{\mathfrak{f}}| = 144,$$

so $[K_{12} : \mathbb{Q}] = 2 \cdot 144 = 288$. The moduli $|v_{12}(k)|^2$ are *real*, and they generate the index-2 real subfield, of degree 16, of the conductor-$(12)\,\infty_1\infty_2$ ray class field of degree 32, which carries i. That subfield is fixed by complex conjugation at the fiducial embedding but is *not* totally real: its signature is $(8, 4)$. The individual moduli split by index: N_0, N_3, N_6, N_9 have minimal polynomials of degree 4 over $\mathbb{Q}$ and the remaining eight have degree 8. Two independent primitive elements, $\sum_k (k+1)\,|v_{12}(k)|^2$ and $\sum_k (2k+1)\,|v_{12}(k)|^2$, each have minimal polynomial of degree exactly 16, fixing the moduli field. As §7.4 shows, the full coordinates generate a proper extension of K_{12}.

7.3 Explicit tower decomposition

The class field decomposes (via `bnrclassfield`) as a compositum of six cyclic pieces over F_{12}, each with a small defining polynomial; the six pieces are collected in Table 1. Four

Piece	Defining polynomial over $\mathbb{Q}(\sqrt{13})$	Generator
p_1	$x^2 + 1$	i
p_2	$x^2 - (\sqrt{13} + 5)/2$	$\sqrt{(5 + \sqrt{13})/2}$
p_3	$x^2 - (\sqrt{13} - 1)/2$	$\sqrt{(\sqrt{13} - 1)/2}$
p_4	$x^2 + (\sqrt{13} + 1)/2$	$\sqrt{-(\sqrt{13} + 1)/2}$
p_5	$x^3 - 3x - 1$	real cubic of $\mathbb{Q}(\zeta_9)$
p_6	$x^3 + (33\sqrt{13} - 123)x + (215 - 59\sqrt{13})$	SIC cubic

Table 1: Explicit tower decomposition of the class field into six cyclic pieces over $\mathbb{Q}(\sqrt{13})$.

quadratic layers and two cyclic cubics give $2^5 \cdot 3^2 = 288$. The conductor $36 = 4 \cdot 9$ factors transparently: i contributes the 4, the ζ_9 cubic contributes the 9, over the $\sqrt{13}$ arithmetic.

7.4 The coordinate field is a ramified S-unit double cover

The construction $v_{12}(k) = \sigma_k(\varepsilon_{12})$ of §3 would place every coordinate in K_{12}, hence at degree dividing 288. A high-precision check refutes this for the crux coordinates and, in doing so, sharpens the construction. We recovered the fiducial to 8000 significant digits and tested each coordinate by a *deep-vanish floor* criterion: a genuine minimal polynomial vanishes to full precision and then plateaus in degree, while a low-precision artifact creeps without flooring. We find $v_{12}(1)$ irreducible of degree $64 = 2^6$. Since $2^6 \nmid 288 = 2^5 \cdot 3^2$, $v_{12}(1)$ *does not lie in* K_{12}; the twelve coordinates generate a proper extension.

The structure is a ramified quadratic double cover. Write $v_{12}(k) = |v_{12}(k)|\, u_k$ with unit phase $u_k = v_{12}(k)/|v_{12}(k)|$. The phase u_1 has degree $32 \mid 288$ and lies in K_{12}, and the square $v_{12}(1)^2 = N_1\, u_1^2$ (where $N_1 = |v_{12}(1)|^2$) is irreducible of degree 32 and also lies in K_{12}. Hence $v_{12}(1) = \sqrt{v_{12}(1)^2}$ is a quadratic extension of a K_{12} element, and the degree 2^6 is precisely one factor of 2 above the 2^5 part of K_{12}: the square root of the modulus.

That square root is a unit at the ramified primes. Modulo squares the ramifying class is $[N_1]$, the phase square u_1^2 dropping out. The minimal polynomial of N_1 over $\mathbb{Q}$ has degree 8, with field norm

$$\mathrm{N}_{\mathbb{Q}}(N_1) \;=\; \frac{9}{9475854336} \;=\; \frac{1}{32448^{\,2}}, \qquad 32448 = 2^6 \cdot 3 \cdot 13^2,$$

a perfect square supported only on the primes $\{2, 3, 13\}$ ramified in K_{12} (field discriminant $2^{12}\, 3^2\, 13^4$). Thus $K_{12}(\sqrt{N_1})/K_{12}$ ramifies only where K_{12} already ramifies: $[N_1]$ is an S-unit class with $S = \{2, 3, 13\}$, and

$$v_{12}(1) \;=\; \sqrt{u} \,\cdot\, (\text{element of } K_{12}), \qquad u \text{ an } S\text{-unit}.$$

The exact-embedding construction of §3 is therefore refined to $v_{12}(k) = \sigma_k(\sqrt{\varepsilon})$ for an S-unit ε: the coordinate is the embedding of the *square root* of the Stark datum, not the datum itself, which resolves the degree count $64 = 2 \cdot 32$. This is the arithmetic shadow of the Frobenius self-pairing $\mu \circ \delta = \mathrm{id}$ of §4: the modulus $N_k = v_{12}(k)\,\overline{v_{12}(k)}$ is the idempotent collapse of a coordinate against its mirror into the real subfield, and the coordinate is that collapse's square-root double cover. The base $F_{12} = \mathbb{Q}(\sqrt{13})$ has fundamental unit $\varepsilon_{13} = (3 + \sqrt{13})/2$ (regulator 1.19476 ...); the exact representative is pinned in §7.5, where the whole coordinate system is presented as one finite ring.

7.5 The existence ring

The double cover says where the coordinates live; the finished construction presents that home as a single finite object. Let $K_{16} = \mathbb{Q}[g]/(p_r)$ be the degree-16 moduli field (real at the fiducial embedding, signature $(8, 4)$), with $p_r = x^{16} - 10x^{14} + 40x^{12} - 90x^{10} + 126x^8 - 96x^6 + 25x^4 + 2x^2 + 1$, in which all twelve squared moduli $N_k = |v_{12}(k)|^2$ are exact rational vectors. Over K_{16} we adjoin four magnitude square roots s_0, s_1, s_3, s_9 (one per independent cover class of §7.4), the imaginary unit i, one real quadratic layer c_5, and one phase generator u_1. The result is a commutative ring with involution

$$R \;=\; K_{16}(s_0, s_1, s_3, s_9,\; i,\; c_5,\; u_1), \qquad \dim_{\mathbb{Q}} R \;=\; 2048.$$

Two collapse relations, found by an exhaustive branch audit of the formal engine against the 1500-digit witness, cut the tower to this size. First, the phase u_1 is *quadratic* over $K_{16}(i)$: its square is the half-angle datum $(c_2 + i\,s_2)/2$ with $c_2, s_2 \in K_{16}$, so the expected octic phase layer is four times smaller than its naive degree count. Second, the fifth cover collapses into the c_5 layer through the exact relation $N_1 N_5 (A_5^2 - 4B_5) = \rho^2$ with $\rho \in K_{16}$, which also places $\sqrt{3}$ and ζ_{12} inside $K_{16}(c_5, i)$.

All twelve coordinates are named elements of R, and the SIC conditions hold in R *exactly*: the norm identity $\sum_k N_k = 1$, the coordinate moduli $\bar{z}_k z_k = N_k$, and all 143 overlap identities $O_{a,b}\,\overline{O_{a,b}} = \frac{1}{13}$ are theorems of `SIC_D12_ExistenceRing.lean`. Each is closed by `native_decide` over exact rational arithmetic (a sparse associative-list engine on 7-bit monomial keys; no floating point appears anywhere in the file). The Lean source is generated by a script that re-executes an independent exact-fraction engine as a gate and emits its vectors verbatim, so the formal data has a single source of truth. Earlier computable models (a quadratic-extension stack for the moduli subfield and a flat degree-288 reduction engine in which θ^{288} reduces under `native_decide`) remain in the development as the route that located this presentation.

The structural payoff of working in R: the identities are ring identities, so *every* star-compatible ring homomorphism $R \to \mathbb{C}$ sends the twelve named elements to a SIC fiducial. Existence in $\mathbb{C}^{12}$ reduces to exhibiting one homomorphism.

7.6 The embedding and the audit

`SIC_D12_Embedding.lean` builds the homomorphism $\varphi : R \to \mathbb{C}$ layer by layer and proves it is a star-ring map on the canonical monomial domain. The base point $g \mapsto g_0$ is the real root of p_r in the rational bracket $(-2.009, -2.008)$, produced by the intermediate value theorem from exact sign changes at rational endpoints (verified by `native_decide` in $\mathbb{Q}$, then lifted to $\mathbb{R}$). The four cover generators map to real square roots of the evaluated moduli; their nonnegativity, the one genuinely analytic obligation, is discharged by exact divided-difference certificates: for each polynomial value, the quotient $q(x) = (p(x) - p(m))/(x - m)$ is computed exactly in $\mathbb{Q}[x]$, giving the bound $p(x) \geq p(m) - \frac{w}{2} \sum_i |q_i| B^i$ on the bracket by pure rational arithmetic. No interval arithmetic and no floating point enter the proof. The layer c_5 maps to the real quadratic root selected by a positive-discriminant certificate, and u_1 is reconstructed by half-angle, $u_1 = \sqrt{(1+x)/2} + i\sqrt{(1-x)/2}$, which makes conjugation compatibility a theorem rather than a branch convention.

The transfer is then mechanical in the best sense. The norm half of the SIC condition is the frozen ring identity $\sum_k N_k = 1$ pushed through φ. For equiangularity, the entire ring side ($|\varphi(O_{a,b})|^2 = \frac{1}{13}$ for all 143 pairs) follows from the `native_decide` identities through the homomorphism; one bridge lemma matches the Weyl–Heisenberg displacement iterates of §2 against the ring overlaps. The bridge pins $Z = \varphi(\zeta)$ by exact computation of its imaginary part ($\operatorname{Im} Z = \frac{1}{2}$) and its twelfth power ($Z^{12} = 1$), so $Z \in \{\omega, \omega^5\}$; both branches are handled by the reindexing $b' = (m\,b) \bmod 12$, a bijection on the 143 nontrivial pairs because $m \in \{1, 5\}$ is a unit modulo 12.

```
theorem d12_sic_exists : IsSICPOVM 12 psi

theorem crystal_forces_d12_sic : SICPOVM_Exists 12
```

The axiom audit closes the account. `#print axioms` on `crystal_forces_d12_sic` reports exactly `propext`, `Classical.choice`, `Quot.sound`, `Lean.ofReduceBool`, and `Lean.trustCompiler`: Lean 4's standard three plus the compiler trust that `native_decide` always carries. Not a single axiom of this development appears in the list; not the Stark axioms of §3, not any shadow assumption. The declaration `crystal_forces_d12_sic` entered the development as an `axiom`; it is now a theorem, and the axiom is deleted.

7.7 Scope

What is established unconditionally: a WH-covariant SIC-POVM exists in dimension 12, by explicit construction, machine-checked from the ring presentation down to the kernel. Exact $d = 12$ fiducials were known in the literature; the formal, audited existence proof is, to our knowledge, the first in any proof assistant, in any dimension. What is established about the arithmetic: the construction confirms the Stark-theoretic predictions in exact form (base field $\mathbb{Q}(\sqrt{13})$, ray class field of conductor $(36)\,\infty_1\infty_2$, coordinates on the ramified S-unit double cover), so the $d = 12$ object is a fully verified instance of the Hilbert-Twelfth construction of §6. What is *not* claimed: no general-d statement follows from the construction; the mixed-signature Stark conjecture is not proved, and the general chain of §3 remains exactly as axiomatized in §3.5.

8 The $d = 2048$ Zauner Transport

Dimension $d = 2048 = 2^{11}$ is the frontier case of the $d = 2^n$ family. Unlike $d = 12$, its unconditional existence is not settled here: it remains the open Zauner conjecture, identified with the mixed-signature Stark conjecture for F_{2048} by Theorem 5.2. What is settled, sorry-free in `ZaunerTransportMap.lean`, is the entire transport apparatus that surrounds that one open node. We prove the unconditional Belnap skeleton, the forward transport and its reduction, the conditional reduction to Stark, and a genuine obstruction to real representations; and we carry the open existence node together with that obstruction as a Belnap B-state rather than resolving it prematurely.

8.1 The problem space and the unconditional skeleton

Write `zauner_d` $= 2048$ and `zauner_n` $= 11$, with the fiducial $B^{\oplus 11}$ the all-B Belnap word of length 11. The structural skeleton is inherited from `sic_povm_belnap_unconditional` of §4, instantiated at $n = 11$ and proved with zero `sorry`s:

$$|O_{B^{\oplus 11}}| = 4^{11} = 4{,}194{,}304,$$
$$\mathrm{frobInner}\big(B^{\oplus 11}, g \cdot B^{\oplus 11}\big) = 22 \quad \text{for every } g \in \mathrm{WHIdx}(11).$$

The orbit has the SIC cardinality d^2, and the join-equiangularity constant $2n = 22$ is uniform across the orbit. No arithmetic hypothesis is used for either statement.

8.2 The transport map and its reduction

The forward transport assigns a complex amplitude to each of the 2048 coordinates from the Hamming weight of its bit vector. For $k \in \mathrm{Fin}\,2048$ with bit vector $v \in \{0,1\}^{11}$ of weight $|v|$,

$$\varphi\big(B^{\oplus 11}\big)\,(k) = \frac{\omega_d^{|v|}}{\sqrt{d}}, \qquad \omega_d = e^{2\pi i/d}.$$

Theorem 8.1 (transport_preserves_norm_sq). *$\varphi(B^{\oplus 11})$ is a unit vector:* `wh_normSq` 2048 $\varphi(B^{\oplus 11}) = 1$.

Proof. Each amplitude has modulus $1/\sqrt{d}$ because ω_d lies on the unit circle and $\|\omega_d^{|v|}\| = 1$. Hence every coordinate contributes $|\cdot|^2 = 1/d$, and the $d = 2048$ coordinates sum to $d \cdot (1/d) = 1$. Proved in Lean 4. □

The reduction $\psi\colon \mathbb{C}^{2048} \to B^{\oplus 11}$ reads a Belnap marginal at each register i by probing two representative coordinates: index 0 (bit i clear) and index 2^i (bit i set). Both carrying amplitude reads B; only the clear probe reads T; only the set probe reads F; neither reads N. This is the structural content of the reverse morphism `AREV`, the read-back half of the split/fuse pair of §4.

Theorem 8.2 (reduction_recovers_fiducial). *$\psi \circ \varphi = \mathrm{id}$ on the fiducial: $\psi\big(\varphi(B^{\oplus 11})\big) = B^{\oplus 11}$.*

Proof. Every transport amplitude is nonzero (a unit phase times $1/\sqrt{d} \neq 0$), so at every register both probes fire and the marginal reads B. The result is the all-B word $B^{\oplus 11}$. Proved in Lean 4. □

Theorems 8.1 and 8.2 exhibit φ and ψ as a transport/reduction pair realizing the Frobenius adjoint half $\mu \circ \delta = \mathrm{id}$ concretely in $\mathbb{C}^{2048}$: the skeleton's structural content survives the round trip into the Hilbert space and back.

8.3 The existence ring and the conditional reduction

The arithmetic geometry mirrors $d = 12$. The SIC discriminant is $m_{2048} = (d-3)(d+1) = 2045 \cdot 2049 = 4{,}190{,}205 = 3 \cdot 5 \cdot 409 \cdot 683$, squarefree, so $F_{2048} = \mathbb{Q}(\sqrt{4190205})$ is real quadratic and the non-square hypothesis holds by `native_decide`. Over it sit the ray class field K_{2048}, the Stark unit ε_{2048}, and the 2048 complex embeddings, all instantiated from §3 at $d = 2048$.

Theorem 8.3 (zauner_transport_theorem)**.** *If the mixed-signature Stark conjecture holds for K_{2048}, then* `SICPOVM_Exists` 2048.

Under the same hypothesis the structural join-equiangularity constant 22 lifts to the metric equiangularity $(d+1)\,|\langle\psi, D_{a,b}\psi\rangle|^2 = 1$, i.e. $|\langle\psi, D_{a,b}\psi\rangle|^2 = 1/2049$, reconciling the discrete skeleton with the $\mathbb{C}^{2048}$ frame (the fuse morphism `FFUSE` of the same pair).

8.4 The character obstruction

The reduction to Stark is the *true* branch. The *false* branch is a genuine obstruction to staying real. We prove it exactly, and state its scope precisely.

Theorem 8.4 (evalf_character_obstruction)**.** *For $d = 2^n$ with $n > 1$, the Weyl–Heisenberg phase character $\omega_d = e^{2\pi i/d}$ is not real: there is no $r \in \mathbb{R}$ with $\omega_d = r$.*

Proof. $\omega_d = \exp((2\pi/d)\,i)$ has imaginary part $\operatorname{Im}\omega_d = \sin(2\pi/d)$. For $d = 2^n > 2$ we have $0 < 2\pi/d < \pi$, so $\sin(2\pi/d) > 0$ and $\omega_d \notin \mathbb{R}$. Proved in Lean 4. □

The content is that the displacement phase is irreducibly complex: a real-valued character of the acting group cannot carry it, so the four-valued false branch (a purely real representation of the WH cocycle) is obstructed, and the projective representation requires genuinely complex cohomology. We are deliberately precise about what this is and is not. It is the exact non-reality of the phase, which is what blocks the real branch. It is *not* the stronger claim that no real fiducial can generate a WH SIC in these dimensions; that statement is adjacent to the real-equiangular-lines bound and we do not assert it. Stating the true obstruction and declining the unproven one is the honest form of the false branch.

8.5 The dialetheic B-state

The two branches do not cancel. The true branch delivers existence under Stark; the false branch delivers the character obstruction; and both are theorems.

Theorem 8.5 (dialetheic_fiducial_paradox)**.** *Assume the mixed-signature Stark conjecture for K_{2048}. Then*

$$\underbrace{\texttt{SICPOVM\_Exists}\ 2048}_{\text{true branch}} \quad \wedge \quad \underbrace{\neg\,\exists\, r \in \mathbb{R} : \ \omega_{2048} = r}_{\text{false branch}}.$$

Recall the Belnap fixed point $\neg B = B$ of §4: the value B absorbs negation and holds true and false at once. Theorem 8.5 places the $d = 2048$ fiducial exactly there. The two dimensions the paper closes therefore sit at opposite Belnap values. At $d = 12$ the existence node is discharged and the fiducial reads T. At $d = 2048$ the existence node is open while its obstruction is proved, and the fiducial reads B: not a gap to be apologized for, but the correct four-valued type of an open problem whose skeleton and whose obstruction are both already theorems.

8.6 Proved unconditionally (0 sorries)

The development proves the following results with zero sorries, without any appeal to the axioms that mark open mathematics.

- **Existence of a SIC-POVM in dimension 12** (`crystal_forces_d12_sic`): exact fiducial constructed in the 2048-dimensional ring R, all 143 overlap identities and the norm machine-checked, transferred to $\mathbb{C}^{12}$ along an explicit homomorphism. Depends on no axiom of this development (§7, audit in §7.6).
- The Belnap four-valued lattice: $\neg B = B$, no explosion, Frobenius closure $\mu \circ \delta = \mathrm{id}$ on $\mathbf{B}_4$.
- Product-lattice orbit cardinality: $|O_{\mathbf{B}^{\otimes n}}| = 2^n$ (`whOrbit_card_eq_pow2`).
- The orbit gap: $2^n < 4^n$ for $n \geq 1$ (`product_lattice_orbit_is_insufficient`).
- The group bifurcation: $|\mathtt{WHIdx}(n)| = 4^n$ (`group_bifurcation_lemma`).
- The main equivalence: `hilbert_embedding_equiv_zauner` by `rfl`.
- The Frobenius closure, join-equiangularity, and Born rule in the Belnap metric (`sic_povm_belnap_unconditional`, 22 theorems).
- The conditional existence theorem (`sic_povm_exists_via_stark`): Stark $\Rightarrow$ SIC, from the axioms.
- **The $d = 2048$ transport** (`ZaunerTransportMap`, 0 sorries): the unit-norm forward transport (`transport_preserves_norm_sq`), the reduction with $\psi \circ \varphi = \mathrm{id}$ (`reduction_recovers_fiducial`), the conditional existence (`zauner_transport_theorem`), and the character obstruction that ω_{2048} is non-real (`evalf_character_obstruction`), with the dialetheic B-state conjoining the true and false branches (`dialetheic_fiducial_paradox`), Theorems 8.1–8.5.

8.7 Axiomatized with known mathematical content

Table 2 collects the axioms that carry the open mathematical content: each marks a specific, known gap in the arithmetic geometry rather than a gap in the formalization strategy.

The gap is the translation from the Stark conjecture embedding bounds to the exact equiangularity and norm equalities. This requires the full arithmetic geometry of Galois orbits of Stark units: the machinery of Shimura varieties and special values of Hecke L-functions applied to the specific WH frame setting. It is not a gap in the formalization strategy; it is a gap in mathematics.

None of these axioms enters the $d = 12$ existence theorem. The audit of §7.6 shows that `crystal_forces_d12_sic` depends only on Lean 4's standard foundations and the compiler trust of `native_decide`. The table above is the honest ledger of the *general-d* chain, not of the $d = 12$ result.

8.8 Open

The mixed-signature Stark conjecture for K_d (Axiom 3.3) and the Zauner conjecture for $d = 2^n$, $n \geq 2$ (`ZaunerConjectureAtPow2`) remain open. Theorem 5.2 identifies them as the

Axiom	Mathematical content
`zauner_correspondence`	Galois–Zauner: WH orbit overlaps controlled by the Zauner automorphism acting on the Stark unit (Appleby et al.)
`equiangular_from_stark_axiom`	Stark bounds imply equiangularity: the full arithmetic geometry of Stark units acting on WH frames
`norm_of_normalized_axiom`	Stark bounds imply the normalized fiducial has the correct norm; same arithmetic geometry gap
`stark_equivalence`	SICPOVM_Exists ↔ Stark unit existence (Appleby–Flammia–McConnell–Yard structural link; placeholder in formalization)

Table 2: Honest gap markers: axioms carrying the open mathematical content.

same problem. For $d = 12$ this subsection is empty: every claim about dimension twelve is a machine-checked theorem. For $d = 2048$ exactly one item is open: the unconditional existence `SICPOVM_Exists` 2048. The transport, reduction, conditional reduction to Stark, and character obstruction of §8 are all theorems; what is not proved is the single node they surround.

9 Discussion

The `rfl` proof of Theorem 5.2 is the main structural result. It states that the Belnap multilattice embedding problem and the Zauner conjecture are definitionally identical: not logically equivalent (which would require a proof), but the same proposition.

This identification works because both definitions reduce to `SICPOVM_Exists (`2^n`)`, the existence of a WH(2^n)-covariant fiducial in $\mathbb{C}^{2^n}$. The Belnap multilattice supplies the structural skeleton (all SIC axioms proved in the Belnap metric, orbit size 4^n, Frobenius closure, Born rule); the question of whether that structure can be realized in $\mathbb{C}^{2^n}$ with the standard WH(2^n) action is exactly what the Zauner conjecture asks. One is the algebraic statement, the other the geometric one; the formalization reveals they are syntactically the same.

The Stark link (§3) extends this to number theory. The discriminant formula $m_d = (d-3)(d+1)$ is not a coincidence; it is the field whose Galois theory organizes the WH orbit. The Galois–Zauner axiom 3.4 is the precise statement of the Appleby–Yadsan-Appleby–Zauner observation [1] that the order-3 element of the Galois group (the "Zauner automorphism") controls the SIC overlaps. Combined with the conditional existence theorem, the SIC-POVM problem becomes, formally, a statement about special values of Hecke L-functions over real quadratic fields.

The present formalization does not prove the Zauner conjecture in generality, nor does it settle Hilbert's Twelfth Problem. It does, however, prove one dimension of Zauner's conjecture outright: $d = 12$, by exact construction, machine-checked from the ring presentation down to the kernel (§7). Exact $d = 12$ fiducials were known in the literature; what is new is the formal existence proof, to our knowledge the first for any dimension in any proof assistant, and with it the deletion of the existence axiom from the development. The method deserves a remark of its own: the entire analytic content of the proof was compressed into a finite set of decidable planks (ring identities under `native_decide`, rational sign changes for the root bracket, divided-difference positivity certificates) plus a single bridge lemma, so the boundary between computation and analysis is drawn exactly where the kernel can stand on the computational side. A numerical fiducial became an exact ring presentation, and the ring presentation became a theorem; at no point does the finished proof cite a floating-point quantity.

For the rest, the formalization locates the open content with precision, identifies the three levels as a single chain, and proves everything in the chain that is not open. The remaining axioms are not placeholders for missing proofs; they are the problem.

10 Artifact statement

All modules cited above live in the `p4ramill` library of the `p4rakernel` repository (`https://github.com/umpolungfish/p4rakernel`). Frozen review snapshot: branch `crystalline/manuscripts3-2026-07-07`, tag `crystalline-manuscripts3-v1` (commit `eea2c0c`); `main` continues independently. This manuscript and its companions `witness_vessel` and `chrysopoeia_2048` are frozen in `ig-docs` (`https://github.com/umpolungfish/ig-docs`) on the same branch name and tag.

11 Acknowledgements

The author thanks Harry T. Larson for imparting the importance of catching rising problems and never letting them go [5]. The present formalization is written in that spirit: the Stark–Zauner–Hilbert identification was not abandoned at the honest gap markers, and dimension $d = 12$ was carried from arithmetic prediction to a machine-checked existence theorem rather than left as a conjectural witness.

A The $d = 12$ fiducial primary coordinate

The machine-checked existence proof of §7 produces an explicit fiducial vector in $\mathbb{C}^{12}$. Its primary (zeroth) coordinate is a complex number whose real and imaginary parts are reproduced below to 1500 significant decimal digits, exactly as recovered by exact rational computation and transferred along the ring homomorphism of §7. This is the numerical witness against which the `native_decide` audit was run.

Real part

```
0.1062796056214234892401116553199052855065537600966583358463645722893226773031270658
298768906205486390530515471062546655918150810594007151620739921064457622344183907390
981527630502926525808781095296318291609375737019796692866006233984692833037108933368
5571418254265468472945125906063607276084935350034065898455161024746050945723981562 11
```

```
32351976687025818044456078364151828066386840393757966879537835285821650394822301169
69793668770812033505406353996844823021336708584735972661333975053476107943755955817
17045877048509213002405037782829689824874235948781256789042674729038344975873958412
89140531346668371523251996235071763355780967913559593392615712936547363385855137100
95980093252520503123502423956964521269530669872896564443756097962597819468821621572
98855888386658756077461233033720151616078844480226265222762050437410169453619512901
48079839035275171184533540544631580441405110469436099763953053419040266213371752782
45334770016486957042075701158765883618692872883510688547764674703272140640514538039
45805914348537665650267990939998246745357874782597423718866832109295985904684538068
58956564454997672734992229740387228401835967003689536235519484175649493189584234922
842411
```

Imaginary part

```
0.00016303100104326943995874063155424964700851818803597898864681891329052651065160
73242458412714148935119300482376377390198732003194349723579628890157953503274150184
94192243297001610296515954002981881435884513376805860796223085630010800409548277559
17676074534782432393712162965636746228269936029409714402315031718519439625197393806
12538628173386359275832075413408173617817250414210122534302281367413780218406525362
81376737724225052724566771677163839774850281518701021145839219108401967581376475783
05921057639967409821600260031107623117350458784774458915383754622457676124458665810
69506278477528999281835417095156724276857660199809330614562037038629851055255370186
61399114025531758967445887190386372014965700841257316611878718834668014275179717067
79479038865240828662979154669327253415544192671945049398602313212212402281654041474
31033338004194588521299511177701905727538444384985519664781080834434965132700067391
42368577763881426844077231076007617110573197696573804943297093786557374673175553241
26243019874514690604566467907604789303836472708176563815896235923819268576683674063
14975286153763506708672554330394678776563753044532036173045952393220064135915855846
64339946520869051498464524476404037246429894784823702974272928188270021773406929319
07879672681199581879502010353314291265458136663163720198654132996034955022563629582
88261801770979857300537231820251970793399646987127813352416035330544430067832933876
15112920627870992379488178912647404950575508868227956502657517079058147927273070715
19045618962335563638457817
```

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